

Abstract

This project evaluates supervised machine learning methods for immune cell type classification using public peripheral blood mononuclear cell single-cell RNA-seq data from CZ CELLxGENE Discover. Four curated immune-cell subset datasets were downloaded, downsampled, merged, and used to construct both broad immune-group and fine-grained cell type classification tasks. Logistic regression was used as the baseline model, random forest as a nonlinear comparison, and sparse logistic regression as an interpretable gene-level model. To reduce data leakage, feature selection, scaling, and PCA were fitted using training data only. Model performance was evaluated using accuracy, macro F1-score, confusion matrices, confidence-threshold analysis, and runtime. Logistic regression achieved strong performance, while random forest did not improve over the linear baseline. Confidence-based rejection allowed uncertain cells to be flagged, and sparse logistic regression identified informative genes associated with predicted cell types. This workflow demonstrates both accurate and interpretable immune-cell classification from processed scRNA-seq data.

1. Introduction

Single-cell RNA sequencing (scRNA-seq) measures gene expression at the level of individual cells, allowing researchers to study cellular heterogeneity that may be hidden in bulk RNA-seq data. This is especially valuable for complex tissues such as blood, where many immune cell types and subtypes coexist in the same sample. A central step in scRNA-seq analysis is cell type annotation, where each cell is assigned a biological identity based on its transcriptomic profile. Accurate annotation is important because downstream analyses, including cell population comparison, disease-state analysis, biomarker discovery, and reference atlas construction, all depend on reliable cell labels. Several computational methods have been developed to support this task, including scmap for reference projection [1], SingleR for reference-based annotation [2], and Garnett for supervised cell atlas annotation [3]. However, benchmarking studies show that annotation performance can vary across methods and datasets, highlighting the need to evaluate models carefully for specific biological classification tasks [4].

Despite these advances, cell type annotation remains challenging. Traditional workflows often cluster cells by gene expression and then assign labels using known marker genes, but this process can be time-consuming, subjective, and dependent on prior biological knowledge. The difficulty increases when cell populations are closely related. For example, broad immune groups such as B cells, NK cells, and T cells are usually easier to distinguish, while fine-grained subtypes such as naive CD4 T cells, naive CD8 T cells, and effector memory CD8 T cells can be harder to separate because they share many T-cell expression patterns. Recent supervised frameworks such as scAnnotatR also emphasize that classifiers should handle uncertain cells rather than forcing every cell into a label [5]. These challenges motivate machine learning approaches that not only classify cell types, but also estimate prediction confidence and provide interpretable features.

Supervised machine learning provides a practical framework for automated cell type classification because it can learn from labeled scRNA-seq data and predict labels for new cells. Prior studies have used logistic regression, random forests, support vector machines, neural

networks, and reference-based methods for this task. Le et al. compared several supervised models for cell type classification from single-nucleus RNA-seq data and found that logistic regression performed strongly for granular cell type prediction [6]. Building on this work, this project evaluates supervised machine learning models for immune-cell classification using processed PBMC scRNA-seq data from the Immunobiology of Aging Cohort PBMC Profiling collection.

Although existing tools show that automated annotation can achieve high performance, several practical gaps remain. Many studies focus mainly on overall accuracy, while less attention is given to the difference between broad immune-lineage classification and fine-grained immune-subtype classification. In addition, standard classifiers often force every cell into a predicted class, even when the model is uncertain. Finally, highly accurate models based on reduced features such as PCA may be less biologically interpretable because they do not directly identify the genes driving each prediction. This project addresses these gaps by comparing broad and fine-grained immune-cell classification, evaluating logistic regression and random forest models, adding confidence-based rejection to flag uncertain predictions, and using sparse logistic regression to identify gene-level features associated with predicted cell types.

Based on these gaps, this project asks whether supervised machine learning models can accurately classify immune cell types from processed single-cell RNA-seq data, and whether model reliability and interpretability can be improved beyond a standard classification baseline. Specifically, the project addresses three research questions: first, how well do logistic regression and random forest models classify broad and fine-grained immune-cell labels; second, can confidence thresholding identify uncertain predictions instead of forcing every cell into a class; and third, can sparse logistic regression provide interpretable gene-level features associated with each predicted cell type. To answer these questions, this project uses public PBMC scRNA-seq data from the Immunobiology of Aging Cohort PBMC Profiling collection. Four curated immune-cell subset datasets were downloaded, downsampled, merged, and used to construct both broad immune-group and fine-grained cell type classification tasks. Logistic regression was used as the main baseline model, random forest was used as a nonlinear comparison model, and sparse logistic regression was used to add interpretability through gene-level coefficients. Model performance was evaluated using accuracy, macro F1-score, per-class precision and recall, confusion matrices, confidence-threshold analysis, and selected gene features from the sparse model.

2. Method

2.1. data acquisition and preprocessing

The dataset used in this project was obtained from CZ CELLxGENE Discover, a public single-cell data portal that provides curated and interoperable single-cell datasets for exploration and download [7]. Specifically, this project used the Immunobiology of Aging Cohort PBMC Profiling collection, which contains peripheral blood mononuclear cell (PBMC) scRNA-seq samples from 234 healthy adults aged 40–89+ collected as part of the Immunobiology of Aging project [8]. Instead of downloading the full all-cells dataset, four curated immune-cell subset files

were selected: Naive CD4 T cells, CD8 T, gdT, MAIT, and dnT cells, NK cells and ILCs, and B and plasma cells. Each subset was inspected using Scanpy to confirm the available metadata, cell type labels, gene features, and expression matrix format. The files were then downsampled, assigned broad labels based on subset identity, and merged using shared gene features. For the fine-grained classification task, the original `cell_type` annotations were retained, and only cell types with sufficient sample counts were kept and balanced by downsampling. Since the downloaded data were already processed, no raw count preprocessing was applied directly. For model training, features were prepared using training-set-only feature selection, scaling, and PCA to reduce data leakage between the training, validation, and test sets.

2.2. Label construction

Two classification label sets were constructed for this project: a broad immune-group label and a fine-grained cell type label. The broad label was created from the curated subset identity of each downloaded CELLxGENE file. Cells from the Naive CD4 T cells subset were assigned the label `Naive_CD4`, cells from the CD8 T, gdT, MAIT, and dnT cells subset were assigned `CD8_related_T`, cells from the NK cells and ILCs subset were assigned `NK_ILC`, and cells from the B and plasma cells subset were assigned `B_plasma`. These labels were stored in a new metadata column called `broad_label` and were used for the broad immune-group classification task.

For the fine-grained classification task, the original `cell_type` annotations provided in the downloaded AnnData files were retained. These labels describe more specific immune-cell subtypes, such as naive B cells, memory B cells, naive CD4 T cells, naive CD8 T cells, effector memory CD8 T cells, and CD16-positive CD56-dim natural killer cells. To avoid unstable evaluation on rare classes, cell types with fewer than 500 cells in the merged dataset were removed. The remaining fine-grained classes were then downsampled to create a balanced dataset, and the retained `cell_type` labels were copied into a new target column called `fine_label`. This produced a more challenging classification task than the broad-label setting while still keeping enough samples per class for reliable model training and evaluation.

This two-level label design allowed the project to compare model performance on an easier broad immune-group task and a harder fine-grained immune-subtype task. It also preserved the original biological annotations while creating simplified labels suitable for supervised machine learning.

2.3 Train/Validation/Test Splitting

After label construction, the datasets were divided into training, validation, and test sets. A 70% / 15% / 15% split was used, where 70% of the cells were used for model training, 15% were used for validation, and 15% were held out for final testing. The validation set was used for model selection, hyperparameter tuning, and confidence-threshold selection, while the test set was reserved for final performance evaluation.

For both the broad immune-group task and the fine-grained cell type task, the split was performed using stratified sampling. This ensured that each class was represented in similar

proportions across the training, validation, and test sets. For the broad-label task, stratification was based on the `broad_label` column. For the fine-grained task, stratification was based on the `fine_label` column. This was important because model performance metrics such as accuracy and macro F1-score can be affected if one split contains more cells from a certain class than another.

The resulting split assignment for each cell was stored in the AnnData metadata as a new column called `split`, with values of `train`, `val`, or `test`. This allowed the same split to be reused consistently across different models and experiments. Keeping the split fixed made it possible to compare logistic regression, random forest, confidence-aware prediction, and sparse logistic regression under the same evaluation conditions.

2.4 Feature Preparation

The downloaded CELLxGENE datasets were already processed and stored as AnnData `.h5ad` files. Initial inspection showed that the expression matrix contained decimal-valued sparse features rather than raw integer counts, and the files also included precomputed embeddings such as PCA, Harmony-corrected PCA, and UMAP. Therefore, raw count preprocessing steps such as normalization and log transformation were not applied again. Instead, the processed expression matrix was used as the starting point for model feature preparation.

For the PCA-based models, including logistic regression and random forest, features were prepared using a training-set-only preprocessing workflow to reduce data leakage. First, highly variable genes were selected using only the training cells. Next, the selected gene features were scaled using a `StandardScaler` fitted only on the training set. PCA was then fitted only on the scaled training data, and the validation and test sets were transformed using the same training-fitted scaler and PCA model. The resulting PCA representation was stored as `X_pca_trainfit` and used as the feature matrix for the logistic regression and random forest models.

For the sparse logistic regression model, the model was trained directly on selected gene expression features instead of PCA components. The same training-set-only feature selection approach was used to select the top variable genes from the training set. These gene features were then scaled using a scaler fitted on the training data and transformed for the validation and test sets. This allowed sparse logistic regression to produce gene-level coefficients while still avoiding feature-selection or scaling information from the validation and test sets.

2.5 Machine Learning Models

Several supervised machine learning models were used to classify immune-cell labels from the processed scRNA-seq features. The main baseline model was logistic regression, trained on the training-fitted PCA representation `X_pca_trainfit`. Logistic regression was chosen as the baseline because it is simple, efficient, and commonly used for high-dimensional biological classification tasks. Since the model learns linear decision boundaries, it also provides a useful

reference point for determining whether the selected cell types are separable in the PCA feature space.

A random forest classifier was used as a nonlinear comparison model. Random forest builds an ensemble of decision trees and can capture nonlinear relationships between features. It was trained on the same `X_pca_trainfit` features as logistic regression so that the two models could be compared under the same input representation and train/validation/test split. This comparison was used to evaluate whether a more complex nonlinear model improved classification performance over the simpler linear baseline.

A confidence-aware logistic regression model was also evaluated. This model used the predicted class probabilities from logistic regression. Instead of always assigning every cell to the class with the highest probability, predictions with a maximum class probability below a chosen confidence threshold were labeled as Uncertain. This allowed the classifier to abstain from low-confidence predictions and provided a way to measure the tradeoff between prediction coverage and reliability.

Finally, a sparse logistic regression model was trained directly on selected gene expression features. This model used L1 regularization, which encourages many coefficients to become zero. As a result, the model can identify a smaller set of informative genes associated with each predicted cell type. Sparse logistic regression was used as an interpretable model to complement the PCA-based models, which may achieve strong performance but are less directly interpretable at the gene level.

2.6 Evaluation Metrics

Model performance was evaluated using several classification metrics. Accuracy was used to measure the overall proportion of correctly classified cells. However, because accuracy can hide poor performance on individual classes, macro F1-score was used as the primary evaluation metric. Macro F1 calculates the F1-score for each class separately and then averages them equally, making it useful for evaluating whether the model performs well across all cell types rather than only the most common or easiest classes. In addition, per-class precision, recall, and F1-score were reported to examine how well each immune-cell class was predicted.

Confusion matrices were also generated for each model to visualize classification errors. These matrices were used to identify which cell types were most often confused with one another, such as closely related T-cell subtypes. This was especially important for the fine-grained classification task, where some immune subtypes are biologically more similar than broad immune-lineage groups.

For confidence-aware prediction, additional metrics were used to evaluate the effect of rejecting uncertain cells. Coverage was defined as the fraction of cells that received a confident prediction, while rejection rate was defined as the fraction of cells labeled as Uncertain. Accepted accuracy and accepted macro F1-score were calculated only on the cells that passed

the confidence threshold. These metrics allowed the project to evaluate the tradeoff between making predictions for more cells and improving reliability among high-confidence predictions.

3. Experiments

3.1 Broad Immune-Group Classification

The first experiment evaluated whether supervised machine learning models could classify cells into broad immune groups. The target label for this task was `broad_label`, which contained four classes: `Naive_CD4`, `CD8_related_T`, `NK_ILC`, and `B_plasma`. This experiment served as an initial baseline to determine whether the selected immune-cell subsets were separable using processed scRNA-seq features. Logistic regression was trained on the training-fitted PCA representation, and model performance was evaluated on the validation and test sets using accuracy, macro F1-score, per-class metrics, and a confusion matrix.

This broad-label experiment was expected to be easier than fine-grained cell type classification because the four classes represent distinct immune groups. Therefore, the purpose of this experiment was not only to measure classification accuracy, but also to establish a reference point for comparison with the more difficult fine-grained classification task.

3.2 Fine-Grained Cell Type Classification

The second experiment used the original detailed `cell_type` annotations to classify more specific immune-cell subtypes. Cell types with fewer than 500 cells were removed to avoid unstable evaluation on rare classes, and the remaining classes were downsampled to create a balanced fine-grained dataset. The final target label was stored as `fine_label`.

This task was designed to be more challenging than broad immune-group classification because some of the retained cell types are biologically related. For example, naive CD4 T cells, naive CD8 T cells, and effector memory CD8 T cells share T-cell expression patterns, making them harder to distinguish than broad lineages such as B cells and NK cells. Logistic regression and random forest models were trained on the same train-fitted PCA features and evaluated using the same train/validation/test split. This allowed a fair comparison between a simple linear model and a nonlinear ensemble model.

3.3 Confidence-Aware Prediction

The third experiment extended the logistic regression model by adding confidence-aware prediction. Logistic regression produces class probabilities for each cell, and the predicted class is normally assigned as the class with the highest probability. In this experiment, the maximum predicted probability was used as a confidence score. If the maximum probability was below a selected threshold, the cell was labeled as `Uncertain` instead of being forced into one of the known cell type classes.

Several confidence thresholds were evaluated on the validation set. For each threshold, coverage, rejection rate, accepted accuracy, and accepted macro F1-score were calculated. Coverage measured the fraction of cells that received a confident prediction, while rejection rate measured the fraction of cells labeled as Uncertain. Accepted accuracy and accepted macro F1-score were calculated only on cells that passed the confidence threshold. Based on the validation results, a threshold of 0.95 was selected because it provided a strong balance between high accepted macro F1-score and high coverage. This threshold was then applied to the held-out test set.

3.4 Sparse Logistic Regression and C Tuning

The fourth experiment trained a sparse logistic regression model directly on selected gene expression features rather than PCA components. This model used L1 regularization to encourage many gene coefficients to become zero, making the model more interpretable by identifying a smaller set of genes associated with each predicted cell type. Unlike the PCA-based logistic regression model, sparse logistic regression allowed the model coefficients to be mapped back to individual genes.

The regularization strength C was tuned using the validation set. Smaller values of C apply stronger regularization and produce a sparser model, while larger values of C allow more genes to contribute to the prediction. The tested values were 0.001, 0.01, 0.05, 0.1, 0.5, and 1.0. The final value of $C = 0.5$ was selected because it provided a practical balance between validation performance, confidence coverage at the 0.95 threshold, and sparsity. After selecting C , the final sparse logistic regression model was evaluated on the test set, and the top positive gene coefficients for each class were extracted for biological interpretation.

3.5 Runtime and Model Comparison

The final experiment compared the models not only by predictive performance, but also by practical tradeoffs such as runtime and interpretability. Runtime was measured by recording training time, prediction time, and total script runtime for each model. This comparison was useful because models with similar accuracy may differ in computational cost and biological interpretability.

The main comparison included logistic regression on PCA features, random forest on PCA features, confidence-aware logistic regression, and sparse logistic regression on selected gene features. Logistic regression served as the primary baseline because it was simple and efficient. Random forest was used to test whether a nonlinear ensemble model improved classification performance. Confidence-aware logistic regression evaluated whether the model could identify uncertain predictions, and sparse logistic regression evaluated whether interpretable gene-level features could be extracted while maintaining strong predictive performance. This experiment helped summarize the tradeoffs between accuracy, reliability, runtime, and interpretability across all models.

Reference

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